Calibration

Provisional revision

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1 Calibration

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The second consists of parameters estimated internally by matching model moments to their empirical counterparts. I describe the two groups in turn, explain which moments are particularly informative about the internally estimated parameters, and then report the model's fit.

1.1 Parameters Set Outside the Estimation

I first describe the parameters assigned outside the estimation. Table 1 summarizes their values and sources.

Demographics and preferences. A model period is four years. Households enter adult life at age 18, retire at age 66, and live at most through age 85. Survival is certain before retirement and thereafter follows four-year probabilities constructed from the combined male and female survivor counts in the 2023 Social Security period life table. Dependent children remain in the household for an expected 18 years. I set risk aversion to $\sigma = 2$.

Within-period utility is CRRA over the Cobb–Douglas composite $c^{\alpha(n)}h^{1-\alpha(n)}$, multiplied by the household equivalence scale $e(n) = ((2 + 0.7n)/2)^{0.7}$. I assign this scale from Scholz et al. (2006), following the Citro–Michael form also used by Borella et al. (2023); the square-root household-size scale is the robustness alternative. Children tilt expenditure toward housing through $\alpha(n) = \alpha_0 - \delta_{\text{jump}}\mathbf{1}\{n \geq 1\} - \delta_a n$. The three expenditure-share parameters are estimated internally. Unlike the circulated specification, the model has no minimum consumption or minimum housing requirement.

Fertility choices are sequential. During fertile ages, a childless household decides whether to attempt a first birth, and a parent decides whether to attempt another birth. Conception succeeds with an age-specific probability that declines to zero at age 45, anchored to Leridon's clinical conception tables. The model distinguishes the taste-shock scale governing entry into parenthood from the scale governing continuation to higher birth orders. Both scales are estimated internally.

Housing finance and transaction costs. Following Greaney et al. (2025), I set the annual real return on liquid wealth to $i_{\text{ann}} = 0.02$, annual housing depreciation to $\delta_{\text{ann}} = 0.011$, the proportional selling cost to $\psi^s = 0.06$, and the labor-income tax rate to $\tau_{\text{pay}} = 0.179$. The return is based on the average ten-year real interest rate, depreciation combines BEA residential-structure depreciation with an adjustment for the land share of house values, and the tax rate comes from OECD data. I set the annual property-tax rate to $\tau^p_{\text{ann}} = 0.01$. Mortgage debt cannot exceed a share $\phi = 0.80$ of the value of the house, so a purchase requires at least 20 percent equity. The benchmark imposes no separate payment-to-income limit.

Income process. I impose a deterministic lifecycle profile of household earnings, normalized to average one over working ages. Within age, log income follows the annual persistent process estimated by Flodén and Lindé (2001), with persistence $\rho_z = 0.9136$ and innovation standard deviation 0.206. I pass this process through the log-linear tax function of Heathcote et al. (2017), using progressivity $\tau = 0.181$. The after-tax process has the same persistence and an innovation standard deviation of 0.169, and is approximated at the model's four-year frequency with a five-state Rouwenhorst chain. Because the equivalence-scale specification has no minimum bundle, this empirical income process can be used without the feasibility restriction that required the circulated specification to truncate income risk.

Housing products and supply. Housing is indivisible and available in a limited assortment of types and sizes (Davis and Van Nieuwerburgh, 2015). I represent this lumpiness with five owner-occupied house sizes, $\{H_k\}_{k=1}^{5} = \{2, 4, 6, 8, 10\}$ rooms. Renters choose housing continuously up to $\bar{h}^R = 6$ rooms, the 90th percentile of the rental-room distribution in the matched ACS sample. These menus summarize the observed tenure-specific distribution of rooms. I set the housing-supply elasticity to $\eta = 1.75$, following the population-weighted metropolitan estimate in Saiz (2010).

Entry wealth and bequests. Households enter at age 18 with a draw from the PSID net-worth-to-income distribution of childless renters aged 18–24. I impose $\theta_n = 0$, so bequest utility does not vary directly with the number of children. Children can nevertheless affect estates through their effects on housing demand, tenure, and saving. Household wealth is measured on the beginning-of-period balance sheet, so liquid wealth and housing refer to the same point in time; bequest flows are measured at the death node after the period's saving decision.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or restriction
Model period	4 years	Model timing
Entry, retirement, and maximum ages	18, 66, and 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security period life table
Expected years with dependent children	18 years	Child-maturity restriction
Risk aversion σ	2	
Equivalence scale $e(n)$	$((2 + 0.7n)/2)^{0.7}$	Scholz et al. (2006); Citro and Michael (1995)
Fecundity by age	Declining; zero from age 45	Leridon anchors; formal fit pending
Family-size states	$0, 1, 2, 3^{+}$; top-bin weight 3.4	Literal child counts; CPS top-bin weight provisional
Annual income persistence ρ_z	0.9136	Flodén and Lindé (2001)
Annual after-tax income innovation s.d. σ_z	0.169	Flodén and Lindé (2001) and Heathcote et al. (2017)
Annual real return i_{ann}	2.0%	Average 10-year real rate, 1962–2024; Greaney et al. (2025)
Annual housing depreciation δ_{ann}	1.1%	BEA structures depreciation, adjusted for the residential land share; Greaney et al. (2025)
Annual property tax τ^{p}_{ann}	1.0%	Maintained tax rate
Financed share ϕ	0.80	20% down-payment restriction; Davis and Van Nieuwerburgh (2015)
Sale cost ψ^{s}	6%	Davis and Van Nieuwerburgh (2015); Greaney et al. (2025)
Labor-income tax τ_{pay}	17.9%	OECD U.S. average
Owner house sizes $\{H_k\}$	$\{2, 4, 6, 8, 10\}$ rooms	Occupied-room distribution in the ACS
Largest rental unit $\bar{h}^{R}$	6 rooms	Matched ACS rental-size distribution
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Wealth-to-income distribution	Childless renters aged 18–24 in the PSID
Bequest child shifter θ_n	0	Maintained restriction

1.2 Internally Estimated Parameters

I estimate the remaining 12 parameters internally. Table 2 reports the estimates from the current diagnostic calibration and the primary empirical counterpart proposed for the revised target system. The parameters are chosen jointly, so the last column is a statement about the intended empirical variation, not a claim that the Jacobian is diagonal. Where one moment cannot separately discipline one parameter, I state an equal-sized parameter–moment block explicitly.

Table 2: Internally estimated parameters and primary empirical counterparts

Parameter	Interpretation	Estimate	Primary empirical counterpart
β_{annual}	Annual discount factor	0.9705	Wealth block: aggregate wealth / gross labor earnings
α_0	Consumption share, no children at home	0.7542	Childless-renter housing-expenditure slope
δ_{jump}	Share shift at entry into parenthood	0.0373	Family-housing block: first-child rooms increment
δ_a	Share shift per additional child	0.0242	Family-housing block: rooms gap, 3+ versus 1–2 children
ψ	Utility from children	2.8137	Childlessness, conditional on first-birth timing
κ_E	Taste-shock scale, entry into parenthood	49.38	Mean age at first birth; absent from the inherited target system
κ_C	Taste-shock scale, continuation decisions	0.1806	Completed fertility, conditional on childlessness
χ_O	Utility premium from owner-occupied housing	1.0588	Ownership rate, ages 30–55
$\bar{H}$	Housing-supply scale	8.5226	Aggregate occupied rooms
θ_0	Strength of the bequest motive	0.0264	Wealth block: annual bequest flow / aggregate wealth
θ_1	Bequest curvature shift	0.0442	Wealth block: living-old wealth p_{90}/p_{50}
κ_T	Dispersion in tenure choices	0.0002	Four-year residual tenure Brier score

Saving. The annual discount factor β_{annual} governs households' willingness to carry resources across ages. Its primary empirical counterpart is aggregate household net worth divided by aggregate gross labor earnings, measured on the same sample and with the same balance-sheet definition in the PSID and the model. For 2005–2019 the target is 6.873. This replaces the borrowed wealth-to-after-tax-earnings value used in the earlier strategy note. The discount factor is still estimated jointly with the bequest parameters, because all three affect lifecycle wealth accumulation.

Housing demand, tenure, and supply. The childless consumption share α_0 has a direct empirical counterpart. For a childless cash renter with Cobb–Douglas preferences, the housing share of expenditure is $1 - \alpha_0$; the CEX housing-expenditure regression therefore implies $\alpha_0 = 0.733$. The two family-housing tilts form a two-parameter, two-moment block. The rooms increment at the first child disciplines the total entry shift $\delta_{\text{jump}} + \delta_a$, while the rooms gap between households with three or more children and those with one or two disciplines the incremental tilt δ_a. This triangular mapping separates the housing adjustment associated with becoming a parent from the additional space associated with larger families.

The remaining three housing parameters have distinct counterparts. χ_O is associated with the ownership rate at ages 30–55, κ_T with residual variation in tenure four years ahead after conditioning on the states represented in the model, and $\bar{H}$ with aggregate occupied rooms. The family ownership gap remains an overidentifying moment: it tests whether the calibrated family-housing tilts interact with the tenure margin as the model intends, but it is not assigned to an additional parameter.

Family size. The three fertility parameters are disciplined by a three-moment block rather than by three independent one-to-one equations. The distribution of first-birth timing, summarized by mean age at first birth, is primarily informative about κ_E. Conditional on that timing margin, childlessness disciplines the entry value of children ψ. Conditional on entry into parenthood, completed fertility and childlessness together imply mean family size among parents, which disciplines the continuation scale κ_C. This mapping is approximately triangular, not exact: ψ affects both entry and continuation, and both shock scales affect completed family size. The current inherited target system contains completed fertility and childlessness but no first-birth timing moment. It therefore does not separately discipline

κ_E, which is why the current diagnostic estimate lies at the boundary of its search region. The share of first births after age 30 and the birth-order progression rates remain validation outcomes.

Bequests. The bequest function is a zero-normalized De Nardi-style luxury warm glow over total bequeathable wealth, including housing. The scale parameter θ_0 governs marginal bequest utility at a given estate and is associated with the annual bequest-flow-to-wealth ratio, 0.0088 in De Nardi and Yang (2014). The shift θ_1 governs how strongly the motive rises across the wealth distribution and is associated with the living-old total-wealth dispersion ratio $p_{90}/p_{50} = 3.448$ measured in the PSID. Thus $(\beta_{\text{annual}}, \theta_0, \theta_1)$ form a three-parameter, three-moment wealth block: aggregate wealth disciplines average accumulation, the bequest flow disciplines the level of resources carried to death, and late-life dispersion disciplines curvature. Each moment loads on all three parameters, but the block no longer asks one late-life wealth level to discipline patience and both bequest parameters simultaneously.

1.3 Target Moments and Fit

The current diagnostic calibration predates the revised mapping above and uses the same 15 moments as the circulated specification. It is reported here to show the quantitative starting point, not as the final implementation of the new target system. Table 3 reports the target moments and their model counterparts.

Table 3: Target moments and model fit (diagnostic calibration)

Moment	Data	Model
Family size and family housing		
Completed fertility	1.918	1.875
Childlessness rate	0.188	0.155
Family ownership gap	0.168	0.072
Housing increment with the first child, rooms	0.664	0.874
Rooms gap, 3+ versus 1–2 children, ages 30–55	0.368	0.554
Tenure and housing consumption		
Ownership rate, ages 30–55	0.575	0.678
Renter mean rooms, no children at home, ages 30–55	3.805	4.348
Owner share with rooms ≥ 6, no children at home, ages 30–55	0.596	0.852
Owner–renter mean rooms gap, no children at home, ages 30–55	2.419	2.038
Ownership rate, ages 25–34	0.341	0.359
Ownership rate, ages 65–75	0.764	0.945
Mean occupied rooms, ages 18–85	5.780	5.937
Wealth and bequests		
Young childless-renter wealth / annual income, ages 25–35	0.179	0.467
Median total estate wealth / annual income, ages 76–84	6.501	6.475
Share with nonhousing wealth $\geq$ annual income, ages 65–75	0.608	0.612

The model closely matches completed fertility, childlessness, young ownership, aggregate occupied rooms, the late-life median estate, and the older nonhousing-wealth share. The sequential fertility specification reaches completed fertility of 1.875 against 1.918 in the data and childlessness of 0.155 against 0.188, two moments that the previous specification could not match jointly.

The largest misses remain in housing and tenure. The model produces an ownership rate of 0.945 at ages 65–75, compared with 0.764 in the ACS, and a family ownership gap of 0.072, compared with

0.168. It also overstates large-home ownership among childless households and the two family housing responses. These outcomes indicate that the inherited target system does not yet balance family housing demand, the ownership margin, and late-life tenure exit.

The wealth moments reveal a related lifecycle tension. Young childless renters accumulate wealth equal to 0.467 of annual income, compared with 0.179 in the PSID. The late-life estate level is matched, but that fit is obtained with the bequest and tenure-dispersion parameters near the lower ends of their search regions. The revised wealth block therefore replaces this combination of young liquid wealth and late-life levels with aggregate wealth, the bequest flow, and late-life dispersion, each measured consistently in the data and the model.

Finally, $\kappa_E = 49.38$ lies at the upper boundary of its search region. With entry into parenthood governed by taste shocks of this scale, fertility is nearly unresponsive to prices and transfers. The current calibration is therefore diagnostic: it establishes that the sequential architecture can match the two completed-family-size moments, but its policy elasticities are not usable until first-birth timing disciplines κ_E and the revised wealth and housing blocks are implemented.

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